

Education

Vellore Institute of Technology

Bachelor of Technology in Computer Science and Engineering

Vellore, Tamil Nadu

August 2023 to May 2027

City Montessori School

ISC Grade 12 — Score: 93.5%

Lucknow, Uttar Pradesh

April 2021 to March 2023

— Commended by the Defense Minister of India for academic excellence; ranked in top 3% nationally.

Technical Skills

- **Languages:** C#, C++, Python, JavaScript/TypeScript, Java, SQL
- **Game Development and Real Time:** Unity 3D, URP, XR Interaction Toolkit, Meta XR SDK, 3D mathematics, transforms, quaternions, physics, collision, WebGL, Three.js, Blender, performance focused systems
- **Backend and Networking:** FastAPI, Node.js, Express.js, RESTful APIs, WebSocket real time pipelines, MongoDB, Redis, PostgreSQL
- **AI and Computer Vision:** OpenCV, MediaPipe, applied computer vision, modern agentic AI coding workflows
- **Foundations and Tools:** Data Structures and Algorithms, Z3 constraint solving, Git and GitHub, Docker, Agile, code reviews, documentation

Experience

AutoVRse

Bengaluru, Karnataka

Software Development Engineer Intern, Enterprise XR

May 2026 to July 2026

- Built WebSocket based real time analytics pipeline streaming session metrics including completion rate and engagement score to React dashboards used by enterprise clients including Tata Group.
- Delivered end to end UI experiences from prototype to production, translating client requirements into deployable interfaces under tight iteration cycles.
- Engineered reusable interaction component libraries for hand tracking, gaze and haptics, reducing module development time by ~40% across client projects while maintaining frame rate targets.

Baldmann

Vellore, Tamil Nadu

Head of Department, Extended Reality

February 2025 to Present

- Lead team of 6 engineers building immersive AR and VR interfaces on Meta Quest 3 with Unity 3D and custom hand tracking systems focused on smooth player experience.
- Pioneered EEG based BCI integration with real time UI biofeedback for adaptive VR environments.

Indian Institute of Technology Delhi

Delhi, India

Research and Software Development Intern

May 2025 to July 2025

- Designed structured data extraction and analysis pipelines leveraging OpenAI API for AI assisted annotation and evaluation.
- Built scalable backend services for multi workflow data processing, improving throughput via parallel ingestion and clear documentation.

Technical Projects

Verse : Multi Game VR Rehabilitation Platform : Unity, C#, WebSocket, Meta Quest 3S

2026

- Engineered a 5 game Unity and Quest platform consisting of Balloon Pop, Garden, Paint Wall, Trophy Forge and Pose with physics based motion, gesture recognition, pose recognition and gaze tracking, unified under one session orchestrator and real time adaptive difficulty engine.
- Implemented zero configuration WebSocket discovery so a companion Flutter clinician app stays live synced with every player session for remote monitoring and control.

Anubhav : VR AI Speech Coach : Unity, C#, WebSocket, Sarvam AI

2026

- **Sarvam AI Track Winner, WE Hack 5.0** — dual channel WebSocket client that streams live audio into a real time STT to LLM to TTS coaching pipeline supporting 22 Indian languages plus English.
- Added a procedural virtual audience that reacts on independent timers, creating believable crowd behaviour and real time feedback for the speaker.

ProductIQ : Physics Validated Product Intelligence : Python, FastAPI, Z3

2026

- Built a FastAPI pipeline that asserts every extracted product specification into a Z3 theorem prover as physics and domain constraints, returning the exact unsat core as a plain language explanation.
- Exposed outlier, duplicate and drift detection across a 12 endpoint REST API, making the system immediately usable by product and data teams.

Hackathons and Awards

12 national hackathon podium finishes, including:

- **Yantra Hackathon 2024** — Best Freshers, Connectron
- **iQOO Hackathon 2026** — First Runner Up, offline on device federated condition monitoring; shortlisted National Grand Finale
- **WE Hack 5.0** — Sarvam AI Track Winner, VR multilingual AI speech coach
- **Cardano Asia Hackathon 2025** — Runner Up, decentralized healthcare dApp; 2nd of 100+ teams at India Blockchain Week
- **Hackulus 2025** — Winner, EMG fatigue analysis with AR and VR plus IoT
- **InnovAct 2025** — Runner Up, XR neurorehabilitation system
- **Sprintathon 2025** — Runner Up, neuromuscular rehabilitation with wireless EMG and IoT; 2nd of 1,000+ teams
- **IInventiv 2025** — Winner, Best Open Innovation, immersive stroke rehabilitation
- **InnoHack 2025** — 3rd Place, assistive smart hand rehabilitation device

National level competitive lawn tennis athlete.